



EVIDENCE GUIDE

Box3D Metal Compatibility and Validation

The exact GPU-resident surface, explicit CPU boundary, differential evidence, and completed acceptance matrix.



Compatibility contract

The reference is unmodified Box3D commit 47d7f7cc7e091142c08d11dc7d2e493c5d34f536. Metal is opt-in and tolerance-equivalent, not bit-identical across platforms.

GPU-resident surface	Modes
Fused integration	Velocity and position across all substeps
Convex contacts	Normal, friction, tangent velocity, twist, rolling, restitution
Mesh contacts	Scalar multi-manifold solve and restitution
Distance joints	Rigid, spring, limit, motor, static bodies
Parallel joints	Soft alignment, torque limiting, static bodies
Overflow	Serial deterministic contact and mixed supported-joint order
Broad phase	Resident leaf/refit plus exact-order raw candidates

Explicit CPU boundary

- Broad-phase topology, pair filtering, narrow phase, and manifolds
- Contact and joint preparation
- Events, islands, sleeping, and CCD
- Recording, queries, topology mutation, and public API calls
- Filter, motor, prismatic, revolute, spherical, weld, and wheel solving
- [Any joint requesting reaction-threshold events](#)

Unsupported constraints keep the reference CPU solve for that step. The position-only Metal stage may still run if its threshold is met.

Differential evidence

Matching CPU and Metal worlds run for multiple steps/substeps. Transforms and linear/angular velocities are compared after each scenario. Values below are observed maxima, not universal bounds.

Scenario	Transform	Velocity
Convex friction	4.77e-7 transform	3.98e-6 velocity
Mesh contacts	4.77e-7	3.60e-6
Distance modes/overflow	1.07e-6	8.27e-5
Mixed distance/parallel	7.45e-9	1.79e-7
Mixed joint overflow	1.82e-6	3.29e-4
Static-body joints	4.47e-8	-

Acceptance matrix

Float and double full suites; CPU-only Release; far-world VF64 containment; AddressSanitizer and UndefinedBehaviorSanitizer; float/double warning-as-error builds; shared dylib demo; install audit; and clean-clone reconstruction.

At (+1e8, -1e8), all 2,048 mixed-shape GPU AABBs contained the same-world CPU oracle. One upload plus ten resident refits was observed across ten contact steps.

Failure semantics. Initialization failure returns false and leaves the world usable. Unsupported work increments fallback counters. Disabling Metal releases resources without destroying the world.